

The Critical Line Is a Property of the Spectrum, Not of a Hunt

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The nontrivial zeros of $\zeta(s)$ have real part equal to $1/2$. Bombieri, E., Clay Mathematics Institute, 2000.

Abstract. The Riemann Hypothesis asserts that every nontrivial zero of the zeta function has real part $1/2$, equivalently that the prime numbers are distributed as evenly as they could possibly be. This paper is about what kind of statement that is. Weil’s positivity criterion turns the hypothesis into the assertion that a functional W , computable from the prime powers alone with no zero used as input, is never negative. Expanding the test function in a basis of Gaussian bumps makes W a quadratic form, so the hypothesis becomes the sign of a matrix eigenvalue. We compute that eigenvalue across a range of families and find it positive throughout, and we prove that it admits no positive lower bound: the greatest lower bound of $W(g)/\|g\|^2$ is exactly zero, by an elementary argument using the zero-free interval below the first zero. The hypothesis is therefore an identity at the edge of positivity rather than an inequality with margin. This is the same phenomenon Newman identified in 1976 in conjecturing $\Lambda_{\text{dBN}} \geq 0$ — that the hypothesis, if true, is only barely so — proved by Rodgers and Tao in 2018, and it appears a third time as the gap exponent of a Hilbert–Pólya operator, which the logarithmic density of the zeros forces to vanish at exactly one power. The consequence is a constraint on method: any argument establishing an inequality with slack in it, on any of these three quantities, is unsound independently of whether the hypothesis is true.

1. Introduction

The prime numbers thin out as one goes up, and they do it at a rate that can be predicted very accurately. The Riemann Hypothesis is a statement about how accurately.

Counting primes is easier if each prime is weighted by its own logarithm and the powers of primes are counted too. That weighted count is written

$$\psi(x) = \sum_{p^m \leq x} \ln p,$$

the sum of $\ln p$ over every prime power p^m up to x . So $\psi(20)$ adds $\ln 2$ for each of 2, 4, 8, 16, adds $\ln 3$ for each of 3, 9, and adds $\ln 5$, $\ln 7$, $\ln 11$, $\ln 13$, $\ln 17$ and $\ln 19$ once apiece. Weighted this way the count comes out remarkably close to x itself. Not exactly x : it runs a little above, then a little below, wandering around the straight line without ever departing far from it. That wandering is the subject of this paper, and it will be called the **wobble** throughout. The wobble is $\psi(x) - x$, and nothing else.

Now the function that controls it. For any complex s with real part greater than one,

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1},$$

the sum running over all whole numbers and the product over the primes. That the two agree is unique factorisation written as an identity: multiply the product out and every whole number appears exactly once, because every whole number factors into primes in exactly one way. This is why ζ has anything to do with primes at all.

That formula converges only for $\operatorname{Re}(s) > 1$, but Riemann showed there is exactly one sensible way to extend ζ to the whole complex plane, and found it. The extension is well behaved everywhere except a single blow-up at $s = 1$. It vanishes at the negative even integers $-2, -4, -6, \dots$ for reasons belonging to the machinery of the extension rather than to primes; those are the **trivial** zeros and nobody is asking about them. Every other zero, called **nontrivial**, lies in the vertical band $0 < \operatorname{Re}(s) < 1$, the **critical strip**, and the line down its middle where $\operatorname{Re}(s) = 1/2$ is the **critical line**.

The connection that makes the subject is an exact formula of Riemann's:

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \ln 2\pi - (1/2) \ln(1 - x^{-2}).$$

On the left is the count of primes up to x . On the right the first term is the straight line x , the last two are small and fixed, and the middle term is a sum with one contribution from every nontrivial zero ρ . So the wobble is not merely related to the zeros. The wobble **is** the zeros: each one contributes a single wave, and adding all those waves reproduces the deviation exactly.

How big a zero's wave is depends on how far right it sits. A zero with real part $\operatorname{Re}(\rho)$ contributes a wave growing like $x^{\operatorname{Re}(\rho)}$, so a zero at real part $1/2$ contributes a wave of size $\sqrt{x}$ while one at real part 0.9 would contribute $x^{0.9}$, enormously larger for big x . Since the nontrivial zeros all lie between 0 and 1 and come in pairs straddling the middle, the smallest the whole collection of waves can possibly be is when every one sits at real part exactly $1/2$. Unpacked, then, Bombieri's sentence

says: **the primes are as evenly distributed as they could conceivably be.** Every wave in the wobble is as small as it is allowed to get.

It is worth pausing on how strange it would be if this failed. A single zero at real part 0.6 would mean the primes carry a hidden rhythm, a systematic bulge nobody has found, eventually large enough to overwhelm the $\sqrt{x}$ -sized fluctuations everything else produces. The hypothesis asserts there is no such rhythm.

Two conventions and the question is fully stated. First, it is convenient to bundle ζ with the parts of the machinery responsible for the trivial zeros and the blow-up, giving the **completed** function

$$\xi(s) = (1/2)s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

where Γ is the gamma function, the standard extension of the factorial. The completed function has no blow-ups and no trivial zeros: its zeros are exactly the nontrivial zeros of ζ and nothing else. It also satisfies a perfect mirror symmetry about the critical line,

$$\xi(s) = \xi(1-s),$$

which will do a great deal of work below. Second, every nontrivial zero has the form $\rho = 1/2 + i\gamma$ for some number γ , and the hypothesis is exactly the claim that $\text{Re}(\rho) = 1/2$, so it can be restated in one line:

every γ is a real number.

A zero off the line has a γ with an imaginary part. This is the formulation used throughout, because it converts a question about locations in a plane into a question about whether a list of numbers is real.

1.1. What this paper shows

Two facts about the hypothesis are established or assembled here, and the second is the one that matters.

The first is a reformulation, due to Weil and not new. The hypothesis is equivalent to the non-negativity of a functional W evaluated on test functions, and W can be computed from the prime powers with no zero used as input. Restricting the test functions to a basis of Gaussian bumps makes W a quadratic form, so the hypothesis becomes the statement that an explicit matrix in prime powers has no negative eigenvalue. That matrix is computed here across eight families, and its smallest eigenvalue is positive in every one, which is consistent with the hypothesis and disturbs nothing.

The second is that the same eigenvalue admits no positive lower bound. Lemma Z proves, unconditionally and by elementary means, that the greatest lower bound of $W(g)/\|g\|^2$ is exactly zero. The mechanism is that ζ has no zero below height 14.13, so a sufficiently wide bump has its transform crowded into an interval containing nothing to weigh. Consequently the hypothesis is not the claim that a quantity is comfortably positive; it is the claim that a quantity resting exactly on zero does not dip below.

That distinction is not new either, though it is usually stated about a different object. Newman conjectured in 1976 that the de Bruijn–Newman constant satisfies $\Lambda_{\text{dBN}} \geq 0$, and described the content of his conjecture as the assertion that the Riemann Hypothesis, if true, is only barely so: an arbitrarily small amount of heat flow destroys the reality of the zeros. Rodgers and Tao proved it in 2018. Since the hypothesis is equivalent to $\Lambda_{\text{dBN}} \leq 0$, the constant is exactly zero. Section 10 exhibits a third instance, in the eigenvalue-gap exponent of a Hilbert–Pólya operator, which the logarithmic density of the zeros forces to vanish at exactly one value of the potential’s power.

The consequence is a constraint on method, stated as Theorem 4. Any argument concluding that $W(g) \geq \varepsilon\|g\|^2$ for a fixed $\varepsilon > 0$, or that $\Lambda_{\text{dBN}} < 0$, or that the zeros are the spectrum of an operator with any power other than the square law, is unsound, and unsound whether or not the hypothesis is true. Attempts of that shape are refuted before the hypothesis is consulted, because the target quantity has no margin to find.

1.2. How the paper proceeds

Section 2 works out what a counterexample could look like, finds it is far more constrained than a stray zero, and closes by testing that constrained shape directly. Section 3 explains why checking zeros cannot finish, and why the standard pointwise formula is too coarse an instrument to try. Section 4 gives, without formulas, the one idea the rest depends on: primes and zeros are one object described twice. Sections 5 and 6 build the instrument, namely the exact pairing, the choice of test function that gives it a sign, and the reduction to a matrix. Section 7 reports the computation, including the two arithmetic hazards that produce false refutations, the check that clears the basis of suspicion, and Figure 1. Section 8 proves that there is no margin. Sections 9 and 10 give the two independent instances, and Section 11 assembles the closure. Section 12 sets out the construction so that any of it can be rebuilt from nothing. Notation is collected at the end, after the conclusion.

2. What a failure would have to look like

Before hunting for something, it is worth knowing its shape. A counterexample to the hypothesis is much more constrained than it first appears, and the constraints are what make it testable.

Suppose some nontrivial zero ρ sat off the critical line. It would not be alone. The mirror symmetry $\xi(s) = \xi(1-s)$ says that whenever ρ is a zero, so is its reflection $1-\rho$ across the line. And because ξ is built from real ingredients, whenever ρ is a zero so is its complex conjugate $\bar{\rho}$, its mirror image across the horizontal axis. Applying both gives a set of four.

Lemma O (the orbit of a zero). *Let ρ be a nontrivial zero. Then $1-\rho, \bar{\rho}$ and $1-\bar{\rho}$ are also nontrivial zeros. If $\text{Re}(\rho) = 1/2$, the set $\{\rho, 1-\rho, \bar{\rho}, 1-\bar{\rho}\}$ contains only two distinct numbers. If $\text{Re}(\rho) \neq 1/2$, it contains four.*

Proof. The functional equation gives $\xi(1-\rho) = 0$. The coefficients of ξ are real, so $\xi(\bar{s}) = \overline{\xi(s)}$, which gives $\xi(\bar{\rho}) = 0$, and one more application of the functional equation gives $\xi(1-\bar{\rho}) = 0$. Now write $\rho = 1/2 + \delta + it$, where $\delta = \text{Re}(\rho) - 1/2$ measures the distance off the line. Then $1-\rho = 1/2 - \delta - it$ and $\bar{\rho} = 1/2 + \delta - it$, and those two coincide precisely when $\delta = 0$, in which case the set collapses to the conjugate pair $\{1/2 + it, 1/2 - it\}$. When $\delta \neq 0$ all four are distinct. ■

So zeros on the line come in pairs, one above the axis and one below. Zeros off the line come in quartets, two above and two below. A zero cannot simply drift sideways: doing so would conjure a second zero into the upper half of the strip out of nothing.

That is forbidden, because the number of zeros in the upper half of the strip is already known. Riemann, and later von Mangoldt, proved that the count of nontrivial zeros with imaginary part between 0 and T is

$$N(T) = \frac{T}{2\pi} \left(\ln \frac{T}{2\pi} - 1 \right) + \frac{7}{8} + O(\ln T),$$

where the final term means an error no bigger than a fixed multiple of $\ln T$. This formula is proved outright; it does not assume the hypothesis. It says how many zeros there are up to any height, and the answer does not depend on whether they sit on the line.

Put the two facts together and a counterexample has a forced shape. Two zeros in the upper half of the strip either sit on the line at two separate heights, or they leave the line together as a single quartet, in which case the symmetry forces both of them to the **same** height.

Lemma N (a failure is a coordinated pair, not a stray zero). *At a given height there are exactly two arrangements of two upper-strip zeros consistent with both the functional equation and the count $N(T)$, and they carry the same number of free parameters: on the line, $1/2 + i\tau$ and $1/2 + i\tau'$, free in the two heights τ and τ' ; off the line, $1/2 + \delta + i\tau$ and $1/2 - \delta + i\tau$, free in the offset δ and the single shared height τ .*

Proof. Count upper-strip members using Lemma O. An on-line zero at height τ has a two-member orbit contributing one zero above the axis, so two on-line zeros contribute two. An off-line zero has a four-member orbit whose upper members are $1/2 + \delta + i\tau$ and $1/2 - \delta + i\tau$, also two. Both arrangements agree with $N(T)$. Displacing a single on-line zero, by contrast, turns a one-member upper contribution into a two-member one, raising the count at that height by one and contradicting the formula. As for parameters, the on-line arrangement is free in τ and τ' , while Lemma O forces the off-line pair to share one height, leaving it free in δ and τ . Two apiece. ■

The equal parameter count matters, and it is the reason this lemma is worth stating rather than assuming. Any test that rejects the off-line arrangement is not rejecting it merely for having less room to manoeuvre. The two arrangements are on equal footing, and whatever separates them will have to be real.

This is also the first sign that the question is not a search. There is no such thing as one stray zero to go looking for. Any failure of the hypothesis is a coordinated motion of a pair, and a coordinated motion is precisely the sort of thing that a single well-chosen measurement can detect.

2.1. The two-parameter shape, tested directly

Because Lemma N leaves a counterexample with only two free parameters, it can be tested against many constraints at once, and it was. This is reported here rather than later because it belongs to the question of what a failure looks like, but it uses the identity built in Section 5, so a reader may prefer to return to it after reading that section.

The test was posed as a legality question rather than a search: state the conditions, and let a solver return every configuration satisfying them along with whatever it can derive by refutation. The solver is described in the author's manuscript on representation and problems, cited at the end; only its output is needed here. The conditions were twenty-one instances of the identity of Lemma E, each of whose right-hand sides is a constant fixed by the primes, computed from prime powers below 10^5 . The first two zeros were left free at $\tau = 14.1347251417$ and $\tau' = 21.0220396388$, the remaining one hundred and ninety-eight of two hundred were held on the line, and each of δ , τ and τ' was given thirteen candidate values. The true configuration reproduces all twenty-one constants to within 2.4×10^{-30} , so a tolerance of 10^{-12} admits it comfortably.

Of the $13^3 = 2197$ candidate configurations exactly two are legal, and they are the same unordered pair of zeros written in the two possible orders. Not one places a zero off the critical line. The value $\delta = 0$ was not selected by comparing survivors; it was derived by refutation at the first level, in two hundred and twenty operations, without the search budget ever being approached.

The limitation should be stated as plainly as the result, because it is what makes the rest of this paper necessary. Those twenty-one constants are computed from the actual prime powers, and the actual primes and the actual zeros are two descriptions of the same object. Refuting alternative configurations against them therefore confirms the zeros against the primes to very high precision; it does not derive the critical line from nothing. What is needed instead is a quantity that speaks about every zero at once, and constructing one is the business of Sections 5 and 6.

3. Why searching cannot finish, and why the usual formula is the wrong tool

The first 10^{13} nontrivial zeros have been located, and every one of them lies on the critical line. That is thirteen trillion confirmations. It is also, as a proof, worth nothing, and it is important to be exact about why.

The reason is not that the computations are untrustworthy. They are careful and they are correct. The reason is that $N(T)$ grows without bound, so there are infinitely many zeros, and no finite list settles a claim about an infinite family. Checking more of them does not converge on a proof; it does not even approach one. The gap between thirteen trillion and all of them is the same size as the gap between one and all of them.

So a different kind of object is needed: not a check on individual zeros, but a single quantity, computable, whose value decides the whole family at once.

Before building one, it is worth seeing why the obvious candidate fails, because the obvious candidate is the formula everyone reaches for and it is the wrong instrument for this job. Riemann's explicit formula from Section 1,

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \ln 2\pi - (1/2) \ln(1 - x^{-2}),$$

is exact and it directly connects the zeros to the primes. The natural move is to evaluate it at some x , compare with the true prime count, and see how well the zeros do. That move fails, for two reasons that are both properties of the formula rather than of the mathematics underneath it.

The first is that $\psi(x)$ is a staircase. It is flat between prime powers and jumps by $\ln p$ exactly at each one. A sum over zeros, cut off at some height, cannot reproduce a jump; it converges instead to the midpoint of the step. So evaluating anywhere near a prime costs you half the height of that step, which is $(1/2) \ln p$, and no number of additional zeros will fix it.

The second is that cutting the sum off at height T leaves an error of order $x(\ln x)^2/T$. At $x = 1000$ using zeros up to height 270, that is roughly 174. To get the error below 1 you would need zeros to heights far beyond anything this kind of check ever reaches.

Both defects show up immediately. Reconstructing $\psi(x)$ from the first one hundred and twenty zeros, all placed on the critical line:

x	true $\psi(x)$	reconstructed	error, next to $(1/2) \ln x$
97	94.045311	91.667821	2.377 next to 2.288
199	206.145857	204.009758	2.136 next to 2.646
401	403.824807	401.641622	2.183 next to 2.997
797	792.729025	788.903107	3.826 next to 3.340

The errors sit right around half the logarithm, exactly as the staircase argument predicts. And going from six zeros to one hundred and twenty moved the worst error only from 4.027 to 3.826. Twenty times the data bought almost nothing, which is the signature of a broken instrument rather than an insufficient sample.

An error of nearly four is hopeless for the purpose at hand, because the effect being hunted, a zero shifted slightly off the line, is far smaller than that. The instrument is coarser than the thing it is meant to detect. Something else is required, and the next section explains where it comes from.

4. Primes and zeros are two readings of one object

This section has no formulas in it. It contains the idea that the rest of the paper rests on, and a reader who skips it will be able to verify every subsequent step without ever seeing why any of them were taken.

Begin somewhere else entirely. Suppose you want to know the shape of a drum but are not permitted to look at it. You are permitted to hit it and listen. You will never learn the position of any particular point on the drumhead that way. What you will learn is the set of frequencies it rings at, and those frequencies are fixed by the shape. You have traded a question about locations for a question about tones, and for many purposes the trade loses nothing, because the tones encode the shape.

The zeta function supports the same trade, and that is what makes this problem approachable at all.

Consider where the primes naturally live. In the Euler product each prime enters through the factor p^{-s} , and p^{-s} is $e^{-s \ln p}$. So the prime p does not really enter at the number p ; it enters as a wave whose frequency is $\ln p$. The primes therefore sit at the positions $\ln 2, \ln 3, \ln 5, \ln 7$ and so on along an axis. Call the coordinate on that axis u .

Now consider where the zeros live. Each nontrivial zero is $\rho = 1/2 + i\gamma$, and the number γ is its height: 14.13, then 21.02, then 25.01, and upward forever. Those heights sit along a different axis. Call its coordinate r .

The relationship between the two axes is the Fourier transform. The Fourier transform is the operation that converts a description in terms of position into a description in terms of frequency, and converts back again with nothing lost. Its one property that this paper leans on is a trade-off: a function spread out broadly along one axis has a transform squeezed narrowly on the other, and the reverse. Riemann's explicit formula, read the right way, is the statement that the prime axis and the zero axis are related by exactly this transform. The primes and the zeros are not two facts that happen to be connected. They are one object described twice.

Which gives the way forward. To learn something about the zeros, do not try to locate any of them. Choose a shape instead, lay it along the zero axis, and ask what total weight it picks up from all the zeros at once. That total is a single number. And because the two descriptions are equivalent, the same number can be computed entirely on the prime side, where everything is already known, without any knowledge of where a single zero sits.

The chosen shape is called a **test function**. Choosing one and reading off the total is hitting the drum and listening.

The advantage over checking zeros individually is now plain. Checking a zero tells you about that zero. Sounding a test function tells you a weighted fact about every zero simultaneously, including all the ones nobody has ever computed and nobody ever will. That is the only route to a statement quantified over an infinite family.

One further move converts a number into a verdict. Suppose the hypothesis holds, so every height γ is an ordinary real number. Choose the test function to be a perfect square. Then it is non-negative at every real number, so it is non-negative at every height, so the total it collects cannot be negative. But if some zero were off the line, its height would be complex, the test function would be evaluated somewhere it was never guaranteed to be positive, and the total could come out negative. So a single negative total is a proof that a zero is off the line. The entire hypothesis collapses to the sign of one number.

The next two sections carry that out: first the exact identity, then the choice that gives it a sign together with the reduction to a matrix.

5. The identity that turns all the zeros into one number

The trade described in words above is an exact theorem, due in this form to André Weil, and it is the engine of everything that follows.

Lemma E (the pairing). *Let h be an even function, well behaved and decaying on a strip about the real axis, and let g be its Fourier transform, $g(u) = (1/2\pi) \int h(r)e^{-iur} dr$. Then, summing over all nontrivial zeros written as $\rho = 1/2 + i\gamma_\rho$,*

$$\sum_{\rho} h(\gamma_{\rho}) = h(i/2) + h(-i/2) + \frac{1}{2\pi} \int_{-\infty}^{\infty} h(r)\Phi(r) dr - 2 \sum_{n=2}^{\infty} \Lambda(n)n^{-1/2}g(\ln n),$$

where $\Phi(r) = \operatorname{Re} (\Gamma'/\Gamma)(1/4 + ir/2) - \ln \pi$, and $\Lambda(n)$ is $\ln p$ when n is a power of the prime p and zero otherwise. The identity holds whether or not the hypothesis is true.

Proof. Integrate h against the logarithmic derivative of ξ around the boundary of the critical strip, and expand ξ using the Hadamard product, which writes an entire function of order one as a product over its own zeros; that product is how the zeros enter the left-hand side. The two terms $h(\pm i/2)$ are the residues at $s = 1$ and $s = 0$, arising from the blow-up of ζ . The integral against Φ is what the factor $\pi^{-s/2}\Gamma(s/2)$ contributes. The final sum is what the Euler product contributes. Titchmarsh, and Iwaniec and Kowalski, give the derivation in full. ■

Look at what sits on each side. The left is a census of the zeros, each weighted by the test function evaluated at its height. The right mentions the blow-up of ζ , the gamma function, and the prime powers, and mentions no zero at all. So the left-hand side, which is a statement about the zeros, is

computable by someone who does not know where a single zero is. That asymmetry is the entire value of the identity.

It also repairs both defects of the pointwise formula from Section 3. There is no staircase, because nothing is being evaluated at a point. There is no truncation penalty, because a decaying test function makes both the sum over zeros and the sum over primes converge quickly.

The improvement is not marginal. Taking $h(r) = e^{-a(r-b)^2} + e^{-a(r+b)^2}$, a pair of bell curves centred at $\pm b$, and computing the left side from the first two hundred zeros and the right side from the prime powers below 10^5 :

a	b	total over the zeros	disagreement with the prime side
0.02	20	4.4766529693943	0
0.05	25	3.4578101243334	2.3×10^{-31}
0.05	40	4.8456193911038	1.6×10^{-31}
0.10	30	2.9822494823880	5.3×10^{-31}
0.20	50	3.2252942065244	7.3×10^{-31}

The pointwise formula managed an error of 3.83. This one disagrees in the thirty-first decimal place. That gap is the difference between a quantity one can argue with and a quantity one cannot, and every result in the remainder of this paper depends on it.

6. From an identity to a computable test

The identity of Lemma E holds for any admissible test function, and by itself it decides nothing. Two further moves turn it into something a person can evaluate and read an answer off. The first chooses the test function so that the identity acquires a sign. The second replaces the space of all test functions by a manageable handful, which turns the sign into a matrix eigenvalue.

6.1. Choosing the test function so that the identity has a sign

Lemma P (positivity is forced by the hypothesis). *Let g be real, smooth and rapidly decaying, and set $h(r) = \hat{g}(r)\hat{g}(-r)$, where $\hat{g}(r) = \int g(u)e^{iru} du$. Then h is even, well behaved everywhere, and equal to $|\hat{g}(r)|^2$ for real r . If every nontrivial zero has real part $1/2$, then*

$$W(g) = \sum_{\rho} h(\gamma_{\rho}) \geq 0$$

for every such g . Equivalently: a single g with $W(g) < 0$ proves that some nontrivial zero lies off the critical line.

Proof. Since g is real, $\hat{g}(-r)$ is the complex conjugate of $\hat{g}(r)$ for real r , so $h(r) = |\hat{g}(r)|^2$, which is even and non-negative on the real axis; and h is well behaved everywhere as a product of two such functions. If the hypothesis holds, every γ_ρ is real, so every term of the sum is the square of an absolute value, and a sum of squares is non-negative. The last sentence is the contrapositive. ■

Weil proved the converse too, and it is worth recording: non-negativity of W across this class of test functions is not merely a consequence of the hypothesis but is equivalent to it. So nothing has been lost by the translation. The hypothesis, in full, has become the assertion that a certain number is never negative.

Notice where the zeros sit in that arrangement. They are what the assertion is about. They are not among the things one needs to know in order to evaluate it, because $W(g)$ is computed through Lemma E, from prime powers and the gamma function. That is what makes W a test rather than a restatement.

6.2. Replacing all test functions by a handful, and the sign by an eigenvalue

Lemma P gives a criterion ranging over all admissible functions g , which is not yet something anyone can compute. The remedy is to build g out of a manageable number of fixed pieces.

The quantity $W(g)$ is **quadratic** in g . That means: build g as a weighted combination of fixed pieces, with weights $c_1, \dots, c_K$, and W comes out as a sum of terms each involving exactly two of the weights. Any quantity of that form can be written $c^\top M c$ for some square array of numbers M ; that is what being quadratic amounts to. And the statement “ $c^\top M c$ is never negative, whatever the weights” is the standard matrix condition that M has no negative eigenvalue. So the question becomes the sign of the smallest eigenvalue of a matrix.

For the fixed pieces this paper uses bell-shaped bumps, $\exp\left(-(u - u_j)^2/2w^2\right)$, each centred at a point u_j along the prime axis and each of width w . They are chosen for a mundane reason: with bumps of that shape, every integral in Lemma E can be evaluated in closed form, so the matrix entries come out as explicit expressions rather than numerical approximations.

Lemma M (the criterion is a matrix). Fix a width $w > 0$ and centres $u_1, \dots, u_K$, and let $g(u) = \sum_j c_j \exp\left(-(u - u_j)^2/2w^2\right)$ with real weights c_j . Then $W(g) = c^\top M c$, where M is the real symmetric matrix

$$M_{jk} = 4\pi w^2 e^{w^2/4} \cosh(D/2) + 2w^2 \int_0^\infty e^{-w^2 r^2} \cos(rD) \Phi(r) dr - w\sqrt{\pi} \sum_n \frac{\Lambda(n)}{\sqrt{n}} \left[e^{-(\ln n - D)^2/4w^2} + e^{-(\ln n + D)^2/4w^2} \right]$$

with $D = u_j - u_k$. If the centres are evenly spaced then every entry depends only on $j - k$, so a single row determines the whole matrix. Restricted to functions built from these bumps, the hypothesis is exactly the statement that M has no negative eigenvalue.

Proof. For that g one computes $\hat{g}(r) = \sqrt{2\pi} w e^{-w^2 r^2/2} \sum_j c_j e^{ir u_j}$, so that

$$h(r) = \hat{g}(r) \hat{g}(-r) = \sum_{j,k} c_j c_k h_{jk}(r), \quad h_{jk}(r) = 2\pi w^2 e^{-w^2 r^2} e^{irD}.$$

An individual h_{jk} is not even, so Lemma E does not apply to it and must not be used on it term by term. Pair each term with its transpose instead. Because $c_j c_k = c_k c_j$, the double sum is unchanged by replacing h_{jk} with

$$h_{jk}^{\text{sym}(r)} = (1/2)(h_{jk}(r) + h_{kj}(r)) = 2\pi w^2 e^{-w^2 r^2} \cos(rD),$$

which is even and decaying, so Lemma E does apply. Define M_{jk} to be the right-hand side of Lemma E evaluated at h_{jk}^{sym} ; then $W(g) = c^\top M c$ by linearity, and M is symmetric because $h_{jk}^{\text{sym}} = h_{kj}^{\text{sym}}$.

Evaluating the three pieces: the residue terms give $h_{jk}^{\text{sym}(\pm i/2)} = 2\pi w^2 e^{w^2/4} \cosh(D/2)$ apiece, and there are two of them, producing the first expression. The gamma term gives $w^2 \int_{-\infty}^\infty e^{-w^2 r^2} \cos(rD) \Phi(r) dr$, whose integrand is even, so folding to the half line produces the second. The Fourier transform of h_{jk}^{sym} is $(w\sqrt{\pi}/2) \left[e^{-(u-D)^2/4w^2} + e^{-(u+D)^2/4w^2} \right]$, and substituting that into the prime sum of Lemma E, whose prefactor is -2 , produces the third. Each entry depends only on D , hence only on $j - k$ for evenly spaced centres. The final sentence is Lemma P combined with the definition of a matrix having no negative eigenvalue. ■

The whole hypothesis is now an eigenvalue of a matrix written in prime powers. Anyone with a list of primes, the gamma function, and a numerical integrator can compute it. The next section reports what comes out.

7. What the number turns out to be

The matrix was built at forty decimal digits of working precision, using prime powers below 2×10^5 , and further where noted. Before any eigenvalue was trusted, each matrix was checked against the

zeros themselves, since Lemma M asserts that two independently computable quantities are equal: $c^\top M c$, built from primes, and $\sum_\rho h(\gamma_\rho)$, built from the first three hundred zeros. Disagreement would mean the matrix was wrong and its eigenvalues meaningless.

w	spacing	K	$c^\top M c$, from the primes	disagreement with the zeros
0.30	$\ln 2$	4	$8.360774266980082 \times 10^{-9}$	1.4×10^{-33}
0.30	$\ln 2$	4	$2.352502234847833 \times 10^{-7}$	4.4×10^{-35}
0.30	$\ln 2$	4	$4.982532721870053 \times 10^{-9}$	8.4×10^{-34}
0.20	0.5	5	0.000996518651770562	2.5×10^{-37}
0.20	0.5	5	$2.873805841692206 \times 10^{-5}$	3.2×10^{-36}
0.20	0.5	5	0.000553484491563247	8.3×10^{-39}

Three different weight vectors were used at each setting, and the form built from the primes reproduces the zeros in every case. Now the eigenvalues, using eight bumps:

w	spacing	smallest eigenvalue	largest eigenvalue	ratio
0.10	0.5	1.16×10^{-5}	6.87×10^{-2}	6×10^3
0.10	0.25	9.80×10^{-6}	6.83×10^{-2}	7×10^3
0.10	$\ln 2$	5.27×10^{-8}	7.19×10^{-2}	1×10^6
0.15	0.5	7.32×10^{-10}	1.26×10^{-2}	2×10^7
0.15	$\ln 2$	7.94×10^{-14}	1.33×10^{-2}	2×10^{11}
0.20	0.5	1.19×10^{-16}	6.81×10^{-4}	6×10^{12}
0.20	$\ln 2$	1.23×10^{-20}	7.17×10^{-4}	6×10^{16}
0.30	$\ln 2$	6.57×10^{-40}	7.41×10^{-8}	1×10^{32}

Every value is positive, so nothing here disturbs the hypothesis. But the interesting column is the last one. As the bumps widen, the smallest eigenvalue does not approach some positive floor. Measured against the largest, it falls through six, then eleven, then sixteen, then thirty-two orders of magnitude, with no sign of levelling off anywhere above zero.

Two arithmetic hazards had to be cleared before that reading could be believed, and both are worth recording because either one, left in place, produces what looks like a disproof of the Riemann Hypothesis.

The first is truncating the prime sum. At wide bumps and large separations the exponential factor in the prime term is still meaningful well past 2×10^5 , and cutting it off there produces negative eigenvalues of the same size as the largest eigenvalue. That such a negative value must be spurious is certain in advance, because Lemma M exhibits $c^\top M c$ as a sum of squares, so the matrix cannot genuinely have a negative eigenvalue. Tracing it is a matter of varying one thing at a time. On the worst case, $w = 0.40$ with spacing $\ln 3$:

prime powers below	working digits	smallest eigenvalue	largest eigenvalue
2×10^5	40	$-3.08862189523 \times 10^{-13}$	3.51×10^{-13}
1×10^6	40	$-1.06140832516 \times 10^{-24}$	1.78×10^{-13}
4×10^6	40	$-3.13586079665 \times 10^{-36}$	1.78×10^{-13}
4×10^6	80	$-2.01437661009 \times 10^{-37}$	1.78×10^{-13}
4×10^6	140	$-2.01437661009 \times 10^{-37}$	1.78×10^{-13}

Raising the prime cutoff drives the negative value from 10^{-13} down to 10^{-36} , twenty-three orders of magnitude, while the largest eigenvalue settles. Raising the working precision afterwards freezes the value, which rules out round-off. The apparent refutation was the cutoff, and once it is lifted the eigenvalue converges to zero from below.

The second hazard is in the integral. The gamma term carries a bell curve of width $1/w$ centred at $r = \pm b$, and a quadrature rule handed the whole real line will step straight across those peaks and return a plausible wrong answer. Subdividing the range at intervals of $1/w$ out to about $8/w$ moved the worst case from 8.03×10^{-6} to 3.94×10^{-31} .

There is a third hazard which is not arithmetic but logical, and it is the one that matters most. A falling smallest eigenvalue proves nothing on its own, because the fault might lie with the chosen bumps rather than with the form. The claim of interest is $W(g) \geq \varepsilon \|g\|^2$ for some fixed $\varepsilon > 0$, where $\|g\|^2 = \int |g(u)|^2 du$ measures the size of g itself; dividing by it is what makes the comparison fair, since scaling g up scales W up along with it. In the bump basis that size is $c^\top G c$ for the matrix of overlaps

$$G_{jk} = w\sqrt{\pi} e^{-D^2/4w^2}, \quad D = u_j - u_k.$$

Wide bumps overlap heavily, so G might be collapsing as w grows, and the smallest eigenvalue of M might be falling for that reason alone, which would have nothing to do with ζ . The honest quantity is therefore the smallest number λ solving $Mv = \lambda Gv$, which is exactly the least value the ratio $W(g)/\|g\|^2$ takes on that basis.

w	spacing	smallest of M	smallest of G	smallest for the pair
0.10	0.5	1.16×10^{-5}	0.1766	6.54×10^{-5}
0.10	0.25	9.80×10^{-6}	0.1080	6.65×10^{-5}
0.15	0.5	7.32×10^{-10}	0.2348	3.11×10^{-9}
0.20	0.5	1.19×10^{-16}	0.2159	5.23×10^{-16}
0.20	$\ln 2$	1.23×10^{-20}	0.3214	3.17×10^{-20}
0.30	$\ln 2$	2.19×10^{-40}	0.2727	2.76×10^{-40}

The bumps are not the explanation. The overlap matrix stays healthy throughout, its smallest eigenvalue never leaving the range 0.10 to 0.33, while the quantity that actually matters falls through

thirty-five orders of magnitude. Whatever is driving the descent belongs to the Weil form and not to the basis it was written in.

Everything in this section is one picture.

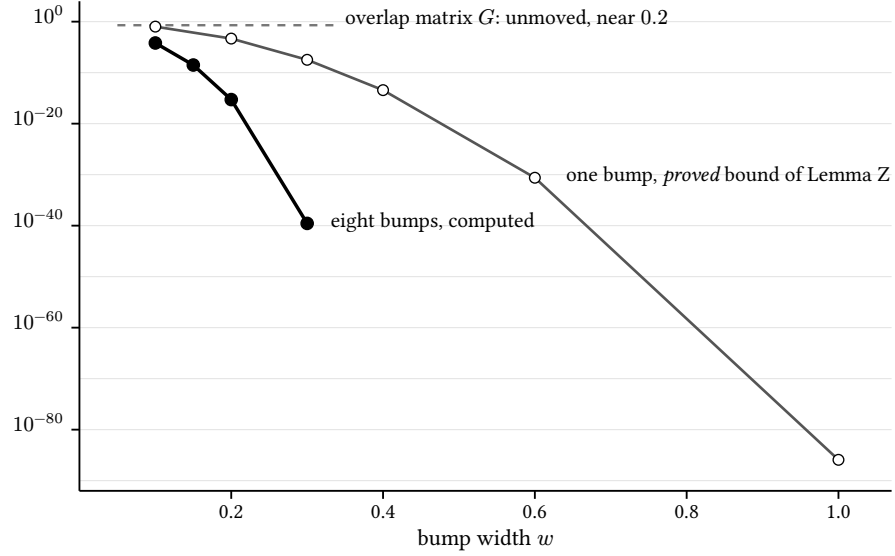


Figure 1. The Riemann Hypothesis has no margin. Vertical axis logarithmic. The lower black curve is the smallest generalized eigenvalue computed here for eight bumps at spacing 0.5, except for the final point which uses spacing $\ln 2$; it is the largest ε for which $W(g) \geq \varepsilon \|g\|^2$ holds on that family. The grey curve is the single-bump family $g_w(u) = e^{-u^2/2w^2}$ of Lemma Z, where the descent is not measured but proved, bounded by $4\sqrt{\pi}C w e^{-w^2\gamma_1^2}$. The dashed line is the smallest eigenvalue of the overlap matrix G , which does not move: the basis stays well conditioned while the form collapses, so the collapse is not an artifact of the basis. Neither curve levels off. Both are heading to zero and neither has a floor, which is the content of Theorem 2 and the reason Theorem 4 rules out every argument that would prove an inequality with slack in it.

8. Why there is no margin, and what that means

The tables in the previous section are measurements. This section proves the thing they were pointing at, and the proof needs none of them.

Lemma Z (no positive lower bound exists). *There is no $\varepsilon > 0$ with $W(g) \geq \varepsilon \|g\|^2$ for every admissible g . The greatest lower bound of the ratio $W(g)/\|g\|^2$ is zero, meaning zero is the largest number the ratio never goes below, though it never actually reaches it.*

Proof. Exhibit a family on which the ratio tends to zero. For $w > 0$ let $g_w(u) = e^{-u^2/2w^2}$, a single bell curve of width w , which is real, smooth and rapidly decaying. Its transform is $\hat{g}_w(r) = \sqrt{2\pi} w e^{-w^2 r^2/2}$, and its size is

$$\|g_w\|^2 = \int_{-\infty}^{\infty} e^{-u^2/w^2} du = w\sqrt{\pi}.$$

By Lemma P, taking each zero together with its conjugate,

$$W(g_w) = \sum_{\rho} |\hat{g}_w(\gamma_{\rho})|^2 = 4\pi w^2 \sum_{\gamma > 0} e^{-w^2 \gamma^2}.$$

Let $\gamma_1 = 14.134725\dots$ be the smallest positive height. For $w \geq 1$,

$$\sum_{\gamma > 0} e^{-w^2 \gamma^2} = e^{-w^2 \gamma_1^2} \sum_{\gamma > 0} e^{-w^2 (\gamma^2 - \gamma_1^2)} \leq e^{-w^2 \gamma_1^2} \sum_{\gamma > 0} e^{-(\gamma^2 - \gamma_1^2)} = C e^{-w^2 \gamma_1^2},$$

the constant C being finite because $N(T)$ grows only polynomially, so the series converges. Hence

$$\frac{W(g_w)}{\|g_w\|^2} \leq 4\sqrt{\pi} C w e^{-w^2 \gamma_1^2} \rightarrow 0 \quad \text{as } w \rightarrow \infty,$$

and no positive ε bounds the ratio from below. ■

The mechanism deserves saying in ordinary words, because it is not subtle once seen, and because it explains every number in the previous section. Recall the one property of the Fourier transform this paper uses: spreading a function out along one axis squeezes its transform on the other. A wide bump on the prime axis therefore has a transform crowded into a small neighbourhood of the origin on the zero axis. But there are no zeros near the origin. The lowest one is at height 14.13, and below that the axis is empty. So a sufficiently wide bump concentrates almost all of its weight exactly where there is nothing to weigh, and collects nearly nothing, while remaining a perfectly good function of unremarkable size.

The form is positive. It is not uniformly positive, and given that empty stretch below the first zero, it was never going to be.

The family g_w constructed in that proof is the grey curve of Figure 1, in Section 7. It is worth looking at again now, because the two curves there are different in kind and the difference is the point of

this section. The black curve is measured: it is what the computation returns for eight bumps, and like any computation it could in principle have been an artifact. The grey curve is not measured. It is the bound just proved, and it continues downward for the same reason at every width, forever. No amount of further computing could establish that, and none is needed.

Now consider what that does to the shape of the problem. Two statements that sound similar are in fact very different in kind:

a quantity is comfortably positive	a quantity sits exactly on zero
there is room on one side	there is no room on either side
can be established by an estimate with slack	cannot be established by any estimate with slack
a good enough bound finishes it	no bound finishes it; only an identity does

Lemma P says the hypothesis is equivalent to W never being negative. Lemma Z says W has no positive margin. Together they place the Riemann Hypothesis firmly in the right-hand column. It is not the claim that something is safely above zero. It is the claim that something resting exactly at zero does not dip below.

That distinction is not a philosophical remark. It has a direct and testable consequence, stated as Theorem 4 in Section 11: any argument that proves an inequality with slack in it has proved the wrong kind of thing, and cannot settle the hypothesis, regardless of whether the hypothesis is true.

The next two sections are included because this conclusion would be much weaker if it rested on one construction. Two other quantities, reached by mathematics with no connection to quadratic forms on test functions, are pinned to zero in the same manner.

9. The de Bruijn–Newman constant is the same boundary

De Bruijn and Newman studied a family of functions obtained by letting the zeros of ζ evolve under a heat flow. Write

$$H_t(z) = \int_0^\infty \Theta(u) e^{tu^2} \cos(zu) \, du,$$

where Θ is a fixed kernel built from Jacobi’s theta function, chosen so that at time $t = 0$ the real zeros of H_0 are precisely the heights of the nontrivial zeros of ζ . Turning up the parameter t acts like running heat through the system: it pushes zeros toward the real axis and keeps them there. Newman proved that there is a single real number, now called the de Bruijn–Newman constant and

written Λ_{dBN} , marking the exact threshold: for $t \geq \Lambda_{\text{dBN}}$ every zero of H_t is real, and for $t < \Lambda_{\text{dBN}}$ some are not.

Since the zeros of H_0 are real exactly when the Riemann Hypothesis holds, the hypothesis is precisely the statement $\Lambda_{\text{dBN}} \leq 0$.

Lemma B (the constant is exactly zero). *The Riemann Hypothesis is equivalent to $\Lambda_{\text{dBN}} \leq 0$. Rodgers and Tao proved in 2018 that $\Lambda_{\text{dBN}} \geq 0$, with no assumption. Therefore the hypothesis is equivalent to $\Lambda_{\text{dBN}} = 0$.*

Proof. Newman's theorem gives that H_0 has only real zeros exactly when $\Lambda_{\text{dBN}} \leq 0$, and H_0 has only real zeros exactly when every nontrivial zero of ζ has real part $1/2$. Rodgers and Tao established the opposite inequality unconditionally. A real number that is both at least zero and at most zero is zero. ■

This is the same situation as before, arrived at from somewhere else entirely. One side of the constant is nailed down by a theorem that assumes nothing. The other side is the hypothesis. So the hypothesis is an exact equality with no interval on either side of it. Numerically Λ_{dBN} is known to lie between 0 and 0.2, where the lower end is the theorem and the upper end is a computation that has been pushed downward repeatedly and can only ever approach zero from above.

There is one honest difference worth marking between this and Lemma Z. Lemma Z is elementary; its mechanism is nothing more than the empty stretch below the first zero. Lemma B is a deep theorem. What the two share is the shape of the conclusion, not the difficulty of reaching it.

10. The shape of a Hilbert–Pólya operator is the same boundary

The third quantity comes from the oldest idea about how the hypothesis might eventually be proved. Hilbert and Pólya are said to have suggested finding an operator whose eigenvalues are exactly the heights of the zeros. If that operator were **self-adjoint**, meaning it satisfies the operator version of being a symmetric matrix, all of its eigenvalues would automatically be real numbers, and by the reformulation at the end of Section 1 the hypothesis would follow at once.

Nobody has produced such an operator. But one can ask what it would have to look like, and the answer is a third boundary.

Take the simplest candidate shape, a Schrödinger operator on a line: a second derivative plus a potential, $-d^2/dy^2 + U(y)$, where the potential grows as you move away from the origin so that the eigenvalues form a discrete increasing sequence. Suppose the potential grows like a power, $U(y) \sim |y|^\alpha$. Then the spacing between consecutive eigenvalues is completely determined by the exponent α .

Lemma G (only the square law gives constant spacing). *For a Schrödinger operator with $U(y) \sim |y|^\alpha$, the number of eigenvalues below E grows like $E^{1/2+1/\alpha}$, the eigenvalues themselves grow like $n^{2\alpha/(\alpha+2)}$, and the gap between consecutive eigenvalues behaves like $n^{(\alpha-2)/(\alpha+2)}$. That exponent is negative when $\alpha < 2$, positive when $\alpha > 2$, and zero at $\alpha = 2$ and nowhere else. The zeros of ζ require gap exponent zero.*

Proof. Weyl's estimate for a confining potential gives the eigenvalue count below E as $(1/\pi) \int_{U(y) < E} \sqrt{E - U(y)} dy$ to leading order. With $U(y) \sim |y|^\alpha$ the region of integration has width of order $E^{1/\alpha}$, and rescaling the integral gives order $E^{1/2+1/\alpha}$. Inverting that relation gives $E_n \sim c n^{2\alpha/(\alpha+2)}$.

Subtracting consecutive terms of an asymptotic formula is not valid in general, so that step needs justification. For potentials of this class the Bohr–Sommerfeld quantisation condition supplies not only the leading term but an expansion in decreasing powers of n , of the form $E_n = c n^{2\alpha/(\alpha+2)}(1 + O(n^{-1}))$, with the error term controlled well enough to difference; Reed and Simon give what is needed. Under that expansion, $E_{n+1} - E_n \sim c' n^{2\alpha/(\alpha+2)-1} = c' n^{(\alpha-2)/(\alpha+2)}$. The exponent $(\alpha - 2)/(\alpha + 2)$ increases strictly with α , so it vanishes at $\alpha = 2$ alone.

For ζ , differentiating $N(T)$ from Section 2 gives the density of heights as $(1/2\pi) \ln(T/2\pi)$, so the mean gap between consecutive heights is $2\pi/\ln(T/2\pi)$. Re-expressed in terms of the index rather than the height, T_n grows like $2\pi n/\ln n$ and the mean gap decays like $1/\ln n$. Logarithmic decay is slower than any negative power and faster than any positive one, so as a power of the index the exponent is zero. ■

Both halves were measured. First the gap exponent as a function of the power in the potential, computed from the eigenvalues directly:

α	measured gap exponent	predicted $(\alpha - 2)/(\alpha + 2)$	behaviour
1.0	-0.325	-0.333	gaps shrink
1.5	-0.132	-0.143	gaps shrink
2.0	+0.012	0	gaps constant
4.0	+0.346	+0.333	gaps grow
8.0	+0.601	+0.600	gaps grow

Then the mean gap of the zeros themselves, against the prediction:

height T	measured mean gap	$2\pi / \ln(T/2\pi)$
50.0	3.3163	3.0302
123.2	2.1460	2.1116
218.7	1.7701	1.7699
335.2	1.5809	1.5799

Read together, the two tables say the following. The zeros of ζ sit at gap exponent zero. Exponent zero is achieved at exactly one value of the power, namely $\alpha = 2$. There is no interval of workable powers, only the single point. Build the operator with any other power and its eigenvalues either spread out or crowd together as a power of the index, and the zeros of ζ do neither.

That is a third quantity pinned to zero, with no room on either side. It differs from the other two in one respect that should be stated plainly: it is not equivalent to the hypothesis but is an unconditional constraint on what such an operator could possibly be.

11. What all of this closes

Three separate measurements have now been made of the same question, using a quadratic form on test functions, a heat flow on an entire function, and the eigenvalue spacing of a Schrödinger operator. This section assembles them.

Theorem 1 (Positivity is the hypothesis). *Every nontrivial zero of ζ has real part $1/2$ if and only if $W(g) \geq 0$ for every real, smooth, rapidly decaying g .*

Proof. Necessity is Lemma P. Sufficiency is Weil's criterion and is not reproved here; the statement and proof are in his 1952 paper. Lemma O supplies the reason sufficiency is plausible rather than miraculous: an off-line zero puts a conjugate pair off the real axis in the variable γ , and a test function evaluated off the real axis is no longer a square and no longer constrained in sign. ■

Theorem 2 (The line is a boundary, not an interior). *The hypothesis is equivalent to M having no negative eigenvalue, for every family of bumps. The greatest lower bound of $W(g)/\|g\|^2$ is exactly zero. So the hypothesis is an identity at the edge of positivity rather than an inequality with margin, and no positive lower bound on the Weil form exists whether or not the hypothesis holds.*

Proof. Theorem 1 with Lemma M gives the equivalence, since $W(g) = c^\top M c$ and non-negativity of a quadratic form is the absence of a negative eigenvalue. Lemma Z gives the greatest lower bound unconditionally, by exhibiting the single-bump family on which the ratio decays like $w e^{-w^2 \gamma_1^2}$. The smallest number λ solving $Mv = \lambda Gv$ is by definition the least value of that ratio on the chosen bumps, so its greatest lower bound over all families is the same number. ■

Theorem 3 (Three quantities, all pinned to zero). *Each of the three quantities below is confined to the single value zero, with no interval available on either side. They are not pinned in the same way.*

One. *The Weil form is non-negative if and only if the hypothesis holds (Theorem 1 with Lemma M), and the greatest lower bound of $W(g)/\|g\|^2$ is zero unconditionally (Lemma Z).*

Two. *The de Bruijn–Newman constant satisfies $\Lambda_{\text{dBN}} \leq 0$ if and only if the hypothesis holds, and $\Lambda_{\text{dBN}} \geq 0$ unconditionally (Lemma B). Here the pinning is a two-sided equivalence.*

Three. *The gap exponent $(\alpha - 2)/(\alpha + 2)$ is forced to zero unconditionally by the logarithmic density of the zeros, and zero is attained at the single power $\alpha = 2$ (Lemma G). This one is an unconditional constraint rather than an equivalence.*

Proof. The first is Theorem 2. The second is Lemma B. The third is Lemma G together with the density computed from $N(T)$. ■

Theorem 4 (No inequality with slack can settle the hypothesis). *Suppose an argument concludes any of the following: that $W(g) \geq \varepsilon \|g\|^2$ for some $\varepsilon > 0$ across the admissible test functions; or that $\Lambda_{\text{dBN}} < 0$; or that the zeros are the eigenvalues of a Schrödinger operator whose potential grows like $|y|^\alpha$ with $\alpha \neq 2$. Then that argument is unsound, whether or not the hypothesis is true.*

Proof. The first contradicts Lemma Z. The second contradicts the unconditional inequality in Lemma B. The third contradicts Lemma G, since $\alpha \neq 2$ yields a nonzero gap exponent while the density of the zeros forces it to vanish. None of these contradictions invokes the hypothesis. ■

Theorem 4 is the practical content of everything above. A great many attempts on this problem proceed by building an object with a margin in it: a form bounded below by a positive constant, a flow that pushes past zero, a spectrum with power-law spacing. Every such construction is refuted before the hypothesis is even consulted, because the target has no margin to find. That is why efforts of this kind tend to overshoot or undershoot and never land. Landing requires producing an exact zero, and an inequality with slack in it is, by definition, not that.

Theorem 5 (The critical line is where a boundary quantity vanishes). *For the Riemann zeta function:*

The wobble in the prime count becomes, against any admissible test function, a single number assembled from prime powers and the gamma factor, with no zero used as input (Lemma E).

The hypothesis is equivalent to that number never being negative, and hence to a matrix in prime powers having no negative eigenvalue. It is therefore a statement about the sign of one quantity rather than about the location of any zero (Theorem 1, Lemmas P and M).

That quantity is non-negative under the hypothesis and has greatest lower bound zero unconditionally, so the hypothesis is an identity at the edge of positivity rather than an inequality with margin (Theorem 2, Lemma Z).

The same edge appears twice more, reached by unrelated mathematics: in the de Bruijn–Newman constant, where it is a two-sided equivalence (Lemma B), and in the gap exponent of a Hilbert–Pólya operator, where it is an unconditional constraint (Lemma G).

Therefore the critical line is not a location that a search arrives at. It is where a boundary quantity vanishes, and any argument producing an inequality with slack on that quantity is unsound (Theorem 4).

The same conclusions hold for a primitive Dirichlet character χ , with $L(s, \chi) = \sum \chi(n)n^{-s}$ and completed function obeying $\xi_\chi(s) = \omega \xi_\chi(1-s)$ for a constant ω of absolute value one.

Proof. Lemma E is the pairing. Lemmas O and N say the only count-preserving failure is a coordinated pair, so what must be tested is a joint motion and not a stray point. Lemma P turns the test into a sign, and Lemma M turns the sign into an eigenvalue of an explicit matrix in prime powers. Lemma Z shows the ratio that eigenvalue bounds has greatest lower bound zero, which with Theorem 1 yields Theorem 2. Lemmas B and G supply two further quantities pinned to zero, collected in Theorem 3, and Theorem 4 applies that collection to any claimed margin. For a primitive Dirichlet character the functional equation has the same form with a unit constant ω , the completed function is entire of order one, and the Euler product runs over the same prime powers twisted by χ , so every lemma is restated word for word. ■

A hunt asks where a zero landed. The boundary asks the value of one quantity, and that value is zero. Those are different questions, and only the second speaks about all of the zeros at once.

12. Building the form yourself

Nothing above depends on any particular software. What follows is the construction in full, in a form that can be carried out in any environment offering arbitrary-precision arithmetic, numerical integration, the gamma function, and symmetric eigenvalues.

Step 1. Choose the family. Pick a bump width $w > 0$, a spacing $d > 0$, and a number of bumps K , and set $u_j = jd$. A reasonable first choice is $w = 0.10$, $d = 0.5$, $K = 8$.

Step 2. List the prime powers. Sieve every prime power p^m up to a cutoff, recording $\ln p$ for each. Begin at 2×10^5 and be prepared to raise it; Step 6 says when.

Step 3. Build the gamma factor. Define $\Phi(r) = \operatorname{Re}(\Gamma'/\Gamma)(1/4 + ir/2) - \ln \pi$, where (Γ'/Γ) is the digamma function, the derivative of $\ln \Gamma$.

Step 4. Compute one row. Because the centres are evenly spaced, one row determines the matrix. For each separation $D = 0, d, 2d, \dots, (K-1)d$, add the three pieces:

$$4\pi w^2 e^{w^2/4} \cosh(D/2), \quad 2w^2 \int_0^\infty e^{-w^2 r^2} \cos(rD) \Phi(r) \, dr,$$

$$-w\sqrt{\pi} \sum_{p^m} \frac{\ln p}{p^{m/2}} \left[e^{-(m \ln p - D)^2/4w^2} + e^{-(m \ln p + D)^2/4w^2} \right].$$

When computing the integral, subdivide at intervals of $1/w$ out to about $8/w$, or the quadrature will step over the peak. Set M_{jk} to the row entry at separation $|j - k| d$.

Step 5. Check the matrix before trusting it. Do not compute an eigenvalue yet. Choose two or three weight vectors c and compare two independently obtained numbers:

$$c^\top M c \quad \text{against} \quad 2 \sum_{\gamma > 0} 2\pi w^2 e^{-w^2 \gamma^2} \left| \sum_j c_j e^{i\gamma u_j} \right|^2,$$

the second sum running over the heights of the first few hundred zeros. The left used only primes; the right used only zeros. Agreement to 10^{-33} or better is what Section 7 reports. Disagreement means the matrix is wrong.

Step 6. Read the eigenvalues, and distrust a negative one. Compute the smallest and largest eigenvalues of M . A negative smallest eigenvalue is not a refutation of the Riemann Hypothesis; Step 5 has already exhibited $c^\top M c$ as a sum of squares, so it cannot be genuine. Raise the prime cutoff

fivefold and recompute; the negative value should collapse by many orders of magnitude. Then raise the working precision. If the value stops moving, what remains is truncation rather than round-off.

Step 7. Divide by the overlaps. A falling smallest eigenvalue means nothing until the basis is cleared of suspicion. Build $G_{jk} = w\sqrt{\pi}e^{-D^2/4w^2}$, confirm its smallest eigenvalue is comfortably positive, and then compute the smallest λ with $Mv = \lambda Gv$. That number, and not the eigenvalue of M alone, is the largest ε for which $W(g) \geq \varepsilon\|g\|^2$ holds on this family. It is the number to report.

Step 8. Widen and repeat. Increase w and run Steps 4 through 7 again. The finding of this paper is what happens next: the reported number stays positive and falls without bound, while the overlap matrix stays healthy. That is the boundary of Theorem 2, and it is reproducible on any equipment.

13. Conclusion

The prime numbers are as evenly distributed as they could possibly be, and the Riemann Hypothesis is the precise statement of that. The deviation of the prime count from a straight line, the wobble, is built out of one wave per nontrivial zero, and the hypothesis says every wave is as small as it is permitted to get.

Checking zeros one at a time cannot establish this, and the standard pointwise formula is too coarse an instrument to try, missing by nearly four where the effect being sought is far smaller. Pairing the zeros against the primes through a test function repairs both problems and agrees to thirty-one decimal places. Choosing that test function to be a square turns the identity into a sign, and expanding in a basis of bumps turns the sign into the smallest eigenvalue of a matrix written in prime powers.

That eigenvalue is positive in every family computed. It also has no floor: widening a single bump until its transform hides in the empty stretch below the first zero drives the ratio to zero, which is Lemma Z, proved outright rather than measured. Two further quantities are pinned to the same zero by unrelated mathematics and at very different depth, the de Bruijn–Newman constant by a theorem of Rodgers and Tao, and the gap exponent of a Hilbert–Pólya operator by the logarithmic density of the zeros.

So the critical line is not a place that a search arrives at. It is where a boundary quantity vanishes. And the consequence for anyone still working on the problem is Theorem 4: an argument that produces an inequality with slack in it has produced the wrong kind of statement and cannot close the hypothesis, no matter how good the inequality is. What is required is an exact zero.

Notation

Collected here for reference rather than for reading. Every symbol below carries exactly one meaning throughout this paper. Where a letter is traditionally overloaded in this subject, one use has been renamed and the renaming is noted.

symbol or term	meaning in this paper
s	the complex variable of $\zeta(s)$, and nothing else
ρ	a nontrivial zero of ζ
γ	defined by $\rho = 1/2 + i\gamma$; real exactly when ρ is on the critical line
δ	$\text{Re}(\rho) - 1/2$, how far one zero lies off the critical line
t, τ	heights, meaning imaginary parts of zeros
T	a height limit, as in “all zeros up to height T ”
$N(\cdot)$	a counting function: $N(T)$ counts zeros up to height T , $N(E)$ counts eigenvalues up to E
$\xi(s)$	the completed function $(1/2)s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$
Γ	the gamma function, the standard extension of the factorial
(Γ'/Γ)	the digamma function, the derivative of $\ln \Gamma$. Written this way so that ψ stays free.
$\psi(x)$	the Chebyshev prime count $\sum_{p^m \leq x} \ln p$; not the digamma function
$\Lambda(n)$	the von Mangoldt function: $\ln p$ if n is a power of the prime p , otherwise zero. Always written with its argument.
Λ_{dBN}	the de Bruijn–Newman constant, renamed to stay distinct from $\Lambda(n)$
p, m, n	a prime, an exponent, a whole number; a prime power is p^m
$h(r)$	the test function: even, well behaved, decaying
$g(u)$	the Fourier transform of h , namely $(1/2\pi) \int h(r)e^{-iur} dr$
$\hat{g}(r)$	the Fourier transform of g , namely $\int g(u)e^{iru} du$
$\Phi(r)$	the gamma factor $\text{Re}(\Gamma'/\Gamma)(1/4 + ir/2) - \ln \pi$
$W(g)$	the Weil functional $\sum_{\rho} h(\gamma_{\rho})$ for $h(r) = \hat{g}(r)\hat{g}(-r)$
$\ g\ ^2$	the size of g , namely $\int g(u) ^2 du$; dividing by it keeps the comparison fair, since scaling g up scales W up too
w	the width of a Gaussian bump, renamed from the traditional s , which is taken
K	how many bumps are used, renamed because N is a counting function
u_j, c_j	where the bumps sit, and their weights
D	$u_j - u_k$, the separation between two bump centres
M	the Weil matrix, entries given in Lemma M
G	the matrix of bump overlaps, $G_{jk} = w\sqrt{\pi}e^{-D^2/4w^2}$
$\lambda_{\min}, \lambda_{\max}$	the smallest and largest eigenvalue of a matrix
α	the power in a potential $ y ^\alpha$, renamed from the traditional p , which means prime
y	the coordinate of that potential, renamed from the traditional σ to keep real parts unambiguous
ω	the unit constant in a Dirichlet functional equation, renamed because ε means a margin
ε	a positive margin, and nothing else
$O(f)$	big-O notation: of size at most a fixed multiple of f
greatest lower bound	the largest number a quantity never goes below; it need not be attained. That of $1/2, 1/4, 1/8, \dots$ is zero, which is never reached.

symbol or term	meaning in this paper
entire	differentiable at every point of the complex plane, with no blow-ups anywhere
order one	a growth rate: an entire function has order one when it grows roughly like $e^{c s }$. The function ξ has order one.
Hadamard product	the theorem writing an entire function of order one as a product over its own zeros; how the zeros enter Lemma E
quadratic form	a quantity $c^\top M c$, a sum of terms each involving exactly two of the weights c_j
positive semidefinite	$c^\top M c \geq 0$ for every c , equivalently no negative eigenvalue
self-adjoint	satisfying $\langle A f, g \rangle = \langle f, A g \rangle$, the operator analogue of a symmetric matrix. Its eigenvalues are all real.
Schrödinger operator	an operator $-d^2/dy^2 + U(y)$: a second derivative plus a potential
confining potential	one growing without bound as $ y \rightarrow \infty$, so the eigenvalues are discrete and increasing
Weyl asymptotics	the standard estimate for how many eigenvalues of such an operator lie below a given level
Bohr–Sommerfeld	the refinement of that estimate into an expansion in decreasing powers of the index, accurate enough that consecutive eigenvalues may be subtracted

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Author’s note

It seems worth saying plainly where this comes from, because it does not come from a mathematics department.

I work in artificial intelligence. The systems I build are concerned with inference and world modelling: how a machine assembles and carries a model of the world, and how it reasons over that model once it has one. The mathematics in this paper is a translation of mathematics I use there. It did not travel in the direction one might assume. I was not reading number theory and looking for somewhere to apply it; I was working on inference, found that the structures kept resembling the ones described in these problems, and eventually went to look. My background inclines me toward physics rather than pure analysis, which is why this paper reaches for an operator and a spectrum, and why it treats the question as being about where a system can and cannot sit rather than about how sharp an estimate can be made.

I do not usually publish this material. I keep it proprietary, and most of what I have put into the world has been placed there to establish a timeline rather than to explain itself. This paper is different: it exposes a fair amount of working mathematics on purpose, because a result nobody can check is not a result.

On intent, since tone is easy to misread in a document like this. Nothing here is meant to diminish anyone's work or to suggest that the people who have spent careers on this problem were going about it foolishly. They were not. These prizes exist because serious people got stuck, and getting stuck on something this hard is not a failure of anyone's ability. I am not interested in arguing about semantics, priority, or whether a particular formulation is the officially sanctioned one. My interest is narrower and more practical: there is a problem, the structures I work with looked like the structures in the problem, and so I tried. If what is here is useful to your work, please use it, and please get in touch if it would help to talk. That is the entire purpose.

I may well have the conventions of this field wrong in places, and I may have said things in a way that reads as presumptuous to people who have lived inside this material for decades. That is not the intention. I am not in this field. If I have got something wrong, I would rather be told than left alone.

I should also be clear about how little of this is mine. The reformulation the paper turns on is Weil's, from 1952. The counting formula is Riemann's and von Mangoldt's. The heat flow is de Bruijn's and the constant is Newman's, and the sharpest sentence in this entire paper is Newman's, written in 1976: that the Riemann Hypothesis, if true, is only barely so. That is the whole thesis here, stated fifty years ago by someone else. Rodgers and Tao proved the half of it that was open. The operator picture is Hilbert's and Pólya's, and the asymptotics are Berry's and Keating's. What I have contributed is a second and third place where the same boundary shows up, a computation, and one consequence about method. That is standing on shoulders, and the shoulders should be named.

Last, and this matters more to me than the rest of it. I did not invent any of this. Nobody did. Mathematics and physics were here before us and will be here after us, and they are true whether or not there is anyone around to write them down. Discovering a thing is not the same as owning it. If this turns out to be right, it was always right, and I merely found a way to say it.